



LAB PRACTICE № 2: EXPRESSIONS, VARIABLES, STATEMENTS, AND OBJECTS IN JAVA

COMP1020: OOP, Algorithms, and Data Structures

Week 02

Lab Practice Submission Instructions:

- This is an individual lab practice and will typically be assigned in the computer laboratory.
- Your program should work correctly on all inputs. If there are any specifications about how the program should be written (or how the output should appear), those specifications should be followed.
- Your classes and methods should be appropriately commented. However, try to avoid making your code overly busy (e.g., include a comment on every line).
- Variables, methods and classes should have meaningful names, and should be followed Java programming convention.
- Academic honesty is required in all work you submit to be graded. Although you can discuss among yourself at high level how to solve the lab problem, you should **NOT** copy or share your code with other students to avoid plagiarism issues.
- Use the template provided to prepare your solutions.
- You should upload your .java file(s) to Canvas **before the end of the laboratory session** unless the instructor gave a specified deadline. No Canvas submission will get 0 point.
- Submit separate *.java file for each Lab problem with the exact file name specified in each lab question (do not add your student ID or your name). For example: **HelloWorld.java**.
- Late submission of lab practice without an approved extension will incur the following penalties:
 - (a) No submission by the deadline will incur 0.25 point deduction for each problem (most of the problems are due at the end of the lab session).
 - (b) The instructor will deduct an additional 0.25 point per problem for each day past the deadline.
 - (c) The penalty will be deducted until the maximum possible score for the lab practice reaches zero (0%) unless otherwise specified by the instructor.

stdin	stdout
16 24	8

Figure 1: Example testcase of Problem 01

Learning Outcome

Topics: Expressions, Variables, operators, assignments, statements, and objects in Java.

Outcome: Acquire the skill of working with the programming language Java basics: evaluating expressions, assigning variables, using operators, calling built-in and module-based functions, using and creating objects.

Problem 01: Greatest common divisor (20 pts)

Write a program and name it Lab02_P01.java: Given 2 positive integers a and b . Compute the greatest common divisor of a and b .

Input

- Line 1 contains 2 integers a, b ($1 \leq a, b \leq 100000$)

Output

- Write the greatest common divisor of a and b .

Example See Figure 1

Hint Pseudo code for finding the greatest common divisor of a and b .

```

1 GCD(a, b){
2     while a != b do {
3         if a > b then a = a - b;
4         else b = b - a;
5     }
6     return a;
7 }
```

Problem 02: Primes (30 pts)

Write a program and name it Lab02_P02.java: Given a positive integer n , list all primes less than or equal to n in an increasing order ($n \leq 10^5$)

Input

- Line 1 contains a positive integers n ($1 \leq n \leq 100000$)

Output

- Write the sequence of primes less than or equal to n in an increasing order (after each prime, write a SPACE character).

Example See Figure 2

Hint Pseudo code for primes listing

stdin	stdout
10	2 3 5 7

Figure 2: Example testcase of Problem 02

```

1  f = new int[100001];
2  for i = 0 to n do f[i] = 0;
3  for i = 2 to n do {
4      if f[i] == 0 then
5          print(i); // i is a prime
6
7      // filter out numbers divisible by i
8      j = i;
9      while j <= n do {
10         f[j] = 1;
11         j = j + i;
12     }
13 }
```

Problem 03: Matrix multiplication (30 pts)

Write a program and name it Lab02_P03.java: Given two matrices $A_{n \times k}$ and $B_{k \times m}$. Each element of the two matrices are integers from 1 to 50. Compute the product matrix $C = A \times B$.

Input

- Line 1 contains two positive integers n and k ($1 \leq n, k \leq 100$)
- Line $i+1$ ($i = 1, 2, \dots, n$): contains the i^{th} row of A (elements are separated by a SPACE character)
- Line $n+2$: contains two positive integers k and m ($1 \leq k, m \leq 100$)
- Line $i+n+2$ ($i = 1, 2, \dots, k$): contains the i^{th} row of B (elements are separated by a SPACE character)

Output

- Line i ($i = 1, \dots, n$) contains the i^{th} row of the resulting matrix (elements are separated by a SPACE character)

Example See Figure 3

Hint Pseudo code of matrix multiplication: $C = A \times B$

```

1  for i = 1 to n do {
2      for j = 1 to m do {
3          C[i,j] = 0;
4          for t = 1 to k do
5              C[i,j] = C[i,j] + A[i,t]*B[t,j];
6      }
```

stdin	stdout
2 3	14 14 14 14
1 2 3	32 32 32 32
4 5 6	
3 4	
1 1 1 1	
2 2 2 2	
3 3 3 3	

Figure 3: Example testcase of Problem 03

stdin	stdout
3	0
5	1
3 2 5 3 4	0
5	
1 2 4 3 5	
5	
5 3 4 0 1	

Figure 4: Example testcase of Problem 04

7 }

Problem 04: Check permutation (20 pts)

Write a program and name it Lab02_P04.java: Given a sequence of n integer $a_1, a_2, \dots, a_n$. Write a program to check if the given sequence is a permutation of $1, 2, \dots, n$

Input

- Line 1: contains a positive integer T ($1 \leq T \leq 100$) which is the number of testcases.
- Subsequent lines are T testcases, each testcase has the following format
 - Line 1: contains a positive integer n ($1 \leq n \leq 100000$)
 - Line 2: contains $a_1, a_2, \dots, a_n$ ($-1000000 \leq a_i \leq 1000000$)

Output

- Line i ($i = 1, 2, \dots, T$) contains 1 if the answer of the i^{th} testcase is YES, and contains 0, otherwise

Example

See Figure 4

Hint permutation is a sequence of n integers from $\{1, \dots, n\}$ in which each integer appears exactly once. Declare an array of boolean $a[1..n]$ in which $a[i] = \text{true}$ to indicate that i appears in the sequence.